

Online Resource 1

Electronic Supplementary Material for

Reference-Conditioned Structural Assessment of Low-Resource Chinese Calligraphy with Overlapping Stroke Parsing and Human-Audited Spatial Scoring

Xiaofan Liu, Ronghao Zhang, and Yuan Feng

Software Engineering Institute, East China Normal University, Shanghai, China

Corresponding author: Xiaofan Liu (10244602411@stu.ecnu.edu.cn)

Journal: *International Journal on Document Analysis and Recognition (IJDAR)*

This Online Resource contains the full controlled-perturbation statistics (Table S1), the perturbation-level alignment ablation (Table S2), the inactive-channel audit (Table S3), the cross-reference test (Table S4), the ASDS candidate search (Note S5), and the integrity hashes of the frozen artifacts (Note S6). All values are computed from the same frozen artifacts as the main text.

Table S1: Full per-perturbation score-drop statistics. Nuisance rows use absolute drop; structural rows use signed drop from the identity score. Confidence intervals use reference-level bootstrap resampling. SD is the standard deviation, P05–P95 the 5th–95th percentiles, Invalid the invalid fraction, ρ_s the mean within-reference Spearman correlation across severity levels, and Monotonic the fraction of adjacent severity steps with non-increasing score.

Perturbation	N	Mean	Median	SD	95% CI	P05–P95	Invalid	ρ_s	Monotonic
Translation	800	0.000	0.000	0.000	[0.000, 0.000]	[0.000, 0.000]	0.000	–	1.000
Rotation	520	5.395	4.890	2.335	[5.079, 5.747]	[3.045, 9.024]	0.133	-0.521	0.647
Scale up	408	13.653	12.775	4.556	[12.964, 14.420]	[8.030, 22.745]	0.490	0.812	0.271
Scale down	800	16.356	15.412	5.225	[15.670, 17.074]	[10.051, 26.419]	0.000	-0.662	0.500
Compound transform	276	8.143	7.298	3.468	[7.621, 8.717]	[4.295, 15.464]	0.540	0.397	0.466
Terminal deletion	800	19.757	11.113	20.424	[17.387, 22.087]	[0.403, 62.702]	0.000	-0.993	0.992
Added direction fragment	800	11.398	6.935	12.736	[9.934, 12.929]	[0.277, 41.357]	0.000	-0.988	0.987
Local fragment shift	800	4.310	2.537	4.220	[3.886, 4.772]	[0.547, 12.677]	0.000	-0.985	0.985
Direction dilation	800	24.034	17.866	18.673	[22.180, 25.955]	[3.336, 62.536]	0.000	-1.000	1.000
Direction erosion	800	22.225	15.425	18.048	[20.269, 24.344]	[3.550, 60.381]	0.000	-0.992	0.997
Endpoint shift	800	7.468	4.570	6.971	[7.294, 7.641]	[0.000, 18.232]	0.000	-1.000	1.000

Table S2: Paired alignment comparisons by perturbation. Positive benefit differences favour the current constrained alignment (Ours); Comparator is no alignment (none) or the wider search (wide); values are mean score drops; RBC is the rank-biserial correlation.

Comparison	Perturbation	Refs	Ours	Comparator	Benefit 95% CI	p	RBC
Ours–none	compound transform	156	8.143	43.947	[34.353, 37.172]	< 0.001	1.000
Ours–none	terminal deletion	200	19.757	2.135	[15.447, 19.802]	< 0.001	1.000
Ours–none	direction dilation	200	24.034	4.624	[17.685, 21.234]	< 0.001	1.000
Ours–none	direction erosion	200	22.225	4.641	[15.636, 19.565]	< 0.001	1.000
Ours–none	added direction fragment	200	11.398	0.923	[9.021, 11.985]	< 0.001	1.000
Ours–none	rotation	180	5.395	15.927	[10.176, 10.871]	< 0.001	1.000
Ours–none	scale down	200	16.356	48.355	[31.504, 32.517]	< 0.001	1.000
Ours–none	scale up	170	13.653	35.869	[20.706, 23.708]	< 0.001	0.988
Ours–none	translation	200	0.000	43.034	[42.054, 44.005]	< 0.001	1.000
Ours–none	endpoint shift	200	7.468	7.468	[0.000, 0.000]	1.000	0.000
Ours–none	local fragment shift	200	4.310	2.270	[1.684, 2.427]	< 0.001	1.000
Ours–wide	compound transform	156	8.143	22.951	[14.336, 15.312]	< 0.001	1.000
Ours–wide	terminal deletion	200	19.757	20.231	[-0.639, -0.317]	< 0.001	-0.696
Ours–wide	direction dilation	200	24.034	24.605	[-0.751, -0.410]	< 0.001	-0.623
Ours–wide	direction erosion	200	22.225	22.730	[-0.682, -0.341]	< 0.001	-0.577
Ours–wide	added direction fragment	200	11.398	11.560	[-0.280, -0.054]	0.005	-0.407
Ours–wide	rotation	180	5.395	12.019	[6.344, 6.908]	< 0.001	1.000
Ours–wide	scale down	200	16.356	16.861	[0.439, 0.571]	< 0.001	0.919
Ours–wide	scale up	170	13.653	14.107	[0.377, 0.533]	< 0.001	0.841
Ours–wide	translation	200	0.000	0.000	[0.000, 0.000]	1.000	0.000
Ours–wide	endpoint shift	200	7.468	7.468	[0.000, 0.000]	1.000	0.000
Ours–wide	local fragment shift	200	4.310	4.302	[-0.006, 0.024]	0.263	0.444

Table S3: Inactive direction channels among the 200 cached references. A channel is inactive when the reference contains no pixel of that direction family; both-empty channels receive Dice 1 under the frozen production convention.

Inactive channels	References	Fraction
0	159	0.795
1	36	0.180
2	5	0.025

Table S4: Cross-reference test of the frozen production score S_{prod} on the external reference library: seven same-character cross-style pairs versus 50 different-character negatives.

Pair type	N	Mean	Median	Mean 95% CI
Same character, cross style	7	20.239	20.272	[15.639, 25.069]
Different-character negative	50	9.741	8.869	[8.584, 10.970]

Note S5: ASDS candidate search

The aligned spatial-distribution similarity (ASDS) was developed after the original 150-pair ratings were available, and those ratings served as its development set. Three fixed distribution families were considered: polar occupancy, 3×3 grid occupancy, and projection profiles. Component weights were evaluated on a coarse non-negative simplex with step 0.025 against the three-rater mean. The full-grid apparent optimum is $(w_g, w_p, w_{\text{proj}}) = (0.225, 0.700, 0.075)$ with Spearman $\rho = 0.557$. Rather than retain the sample-specific secondary weights, the search-selected polar weight was kept and the remaining 0.30 was split equally, giving the frozen specification $(0.15, 0.70, 0.15)$ with $\rho = 0.556$. Five-fold internal validation groups pairs by character identity, selects the weights of each fold on the other characters only, and yields the character-grouped

out-of-fold estimate $\rho = 0.540$ reported in the main text. The individual components reach $\rho = 0.547$ (polar), 0.447 (grid), and 0.445 (projection). All of these are development or internal-validation results on one retrospective cohort; none is an external or prospective confirmation. The search grid and its selected optimum are recorded in the frozen development report so that post-rating researcher degrees of freedom remain visible.

Note S6: integrity hashes of frozen artifacts

The following SHA-256 values identify the frozen contracts and result files from which the manuscript values are computed. The digests were computed on the original files with CRLF line endings; a text file checked out with LF line endings must be re-encoded to CRLF before its digest is compared.

Artifact	SHA-256 (printed as two 32-character halves)
Recovered legacy acquisition archive	b9924007099033cc8b62128dc2139ea9 cb04a66a48e56c46518407677254450d
Dataset QC exclusion contract	bd2b0641d0e6f53f6f18f6604232c02f f99e9d989eb39125f6a9af41e8573a1a
Character-disjoint split (before QC)	eec9bf5c0910a2e9f6046991f1458519 cd903d31deea3e0a4d33c555ff53a09e
QC-clean character-disjoint split	e9303314d1b70d3f92efcdc5c0807f83 3148cbe64c2702379f0ac951ed2a1e2b
Task 1 per-seed result table	c6d27f365fab15b71914c6ae1ccaf723 b02e236c28dc2b4a656ad7c50ea7934e
Task 1 checkpoint manifest	f49c4fc8b364abe62eade730dc6f9b14 e35d6baf809d5d7e5e6fb3d7462b391e
Reference cache index	6d14374fda04de3957b29a874d16cbf9 86ccbc1f1d8be1bcfab8285f0fd9f00d
SegFormer-B2 v1 checkpoint	64df27aafc0eeec07c0ac52c6ff00ee f6b290ae7baf964cd5cf786262f395ce
Frozen human-validation pair file	810d35e4e4bd0208de9608054daabaec c3613859f3a34c5a16e5141e266d8e66
Frozen ASDS development specification	da79ace3dca2793f586ef721689cbb96 e1a0e614155a8c94ff48f6cb7872fa4e
ASDS development report	01fe8d62f368fbc45d5a4451a3d833ed 1d75c7a4896c943e8c8590e790ec3083
Direct-ink ASDS report	44403497fd4980c834946b761b8352a1 b05f49249f14c4a8f668affac36ac895